



Developing a framework for innovating less-structured business processes: a Delphi study

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Abstract

In dynamic business environments, many organizations need to acquire a process innovation capability (PIC) that caters to less-structured business processes (LSBP) in order to remain relevant in the market. However, such organizations are still in need of appropriate frameworks that enable them to identify and evaluate their PIC in LSBP. To fill this gap, we have followed the Delphi method with three rounds of semi-structured interviews, including twelve practitioners and fifteen scholars across five continents. The findings reveal five major differences and barriers for PIC in LSBP, as compared to innovating structured business processes. In addition, we refine and empirically validate an innovation capability framework covering six main capabilities and 18 sub-areas, configured into three major pillars: (1) a people-process-technology pillar, (2) an organizational pillar, and (3) an ecosystem pillar. Besides being one of the first studies to offer a conceptual framework for PIC in LSBP, we provide practical brainstorming with reflective questions per capability, for practitioners to reflect on their organization's current PIC in LSBP.

Keywords Delphi · Framework development · Less-structured business processes · Process innovation

1 Introduction

Organizations operating in dynamic environments need to respond to changes in a quick and flexible way (Reichert and Weber 2012) in order to stay relevant and differentiate themselves. Such uncertain and rapid changes include a variety of possi-

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bilities, such as shifts in customers' attitudes, implementations of new law (Weber et al. 2009), technology disruptions, natural disasters, pandemics, political aftermaths, or terrorist attacks (Bandara et al. 2021). Organizations have therefore recognized the necessity of responding to such external changes by incorporating new ideas into their business processes, and developing their innovation capability for both structured and unstructured business processes (Dumas et al. 2018; Kerpedzhiev et al. 2021). A capability is a nontransferable organization-specific resource that is used to enhance the productivity of the organization's other resources (Makadok 2001). It refers to the ability to recognize the importance of how organizations learn and orchestrate assets in such ways that markets cannot replicate. Innovation capability, and particularly process innovation capability, can help organizations enhance their performance, increase demands, and lower costs (Jansen et al. 2006; Zimmermann, D.F. Ferreira, & Carrizo Moreira, 2016).

Process innovation in well-structured business processes is more conventional because most organizations tend to develop a process model with as much structure as possible by pairing from existing routine to enhance efficiency and standardization (Benner and Tushman 2003). However, the landscape shifts when dealing with less-structured business processes (LSBP), which are characterized by highly dynamic requirements that need knowledge-intensive employees, and for which process innovation is also essential but less evident (vom Brocke et al. 2015). Process innovation in LSBPs faces more challenges for effective decision-making, discussion, and collaboration than the structured business processes (SBP). Nonetheless, LSBPs find applications in diverse scenarios, among others for exception and management escalation, adapting recovery and response assistance for unexpected contextual changes during disasters, managing patient case management in hospitals, and complex project management (Di Ciccio et al. 2015; Kemsley 2011). The complexities of LSBPs stem from their inherent flexibility and dynamics in nature and necessitate differentiating from a conventional approach that is designed for structured environments. It is important to acknowledge this necessity, given that LSBPs require innovative solutions that accommodate uncertainty, foster adaptability, leverage the creative potential of knowledge-intensive employees and varying contextual factors of BPM (vom Brocke et al. 2015). Despite extensive research on process innovation for structured business processes (SBP) and the utilization of certain methods towards LSBP (e.g., case management), previous research that combines process innovation and LSBP is rather limited (Grisold et al. 2021), mainly because it does not fully acknowledge the fact that business processes can have distinct characteristics, necessitating the use of alternative approaches. Therefore, our study will address that need to help organizations develop a process innovation capability (PIC) when dealing with LSBP.

The purpose of this study is to first differentiate innovating LSBP from SBP, and (based on these insights) develop a refined framework with specific criteria for innovating LSBP. Building upon existing related capability frameworks, such as (Goni and Van Looy 2022), our study aims to empirically validate, extend and refine a comprehensive framework that is dedicated to the specific LSBP context, while also turning this conceptual overview of LSBP-specific capabilities into a practical assessment opportunity by deriving related reflective questions. Such practical questions can help organizations brainstorm about their specific business context and enhance

their innovation capabilities in LSBP. Therefore, the following research questions will be addressed in this article:

- RQ1. What are the main differences and barriers between innovating structured and less-structured business processes?
- RQ2. What are the capabilities for innovating less-structured business processes?

Our research questions will be empirically addressed by following a modified Delphi method, which focuses on achieving consensus and gaining judgment on complex issues when precise information to support a decision or prediction is unavailable (Richards and Curran 2002; Sobaih et al. 2012). It allows subject matter experts to address the complex issue of innovating LSBP while treating their level of expertise in a systematic and anonymous manner (Linstone and Turoff 1975; Okoli and Pawlowski 2004; Skutsch and Hall 1973). Our intended research contributions are three-fold, namely: (i) identifying the main differences and barriers between innovating in SBPs and LSBPs, (ii) iteratively building and testing an LSBP innovation capability framework, and (iii) offering brainstorming reflections for practitioners. Besides providing knowledge contributions by extending the literature on process innovation with a validated framework of important criteria, our managerial implications of the proposed framework will enable organizations to translate the uncovered capabilities into a specific business context when innovating their LSBPs.

In [Theoretical background](#), we proceed with the theoretical background on different business process types and innovation capability frameworks. In [Methodology](#) and [Results](#), we respectively describe the chosen method and present the results of the Delphi study. [Discussion](#) continues with discussing our contributions, limitations and directions for future research. Section 6 concludes the study.

2 Theoretical background

We start with defining PIC and LSBP, before presenting the differences between existing innovation capability frameworks.

2.1 Process innovation capability (PIC) and less-structured business processes (LSBP)

Process innovation is defined as “*a new way of making the product or rendering the service*” (Damanpour 2020, p. 22). It refers to the development of a new or significantly changed business process to deliver products and services (Camisón and Villar-López 2014), or to perform a job activity in a radically new way (Davenport 1993). Many organizations acknowledge the necessity of incorporating an innovation capability into their processes to remain competitive (Davenport 1993), with a process innovation capability being defined as “*a firm’s ability to acquire, assimilate, transform, and exploit technically related resources, procedures, and knowledge for process innovation purposes*” (Frishammar et al. 2012, p. 521).

Although process innovation has attracted much less attention than product innovation (Keupp et al. 2012), it can help organizations in two ways. First, process innovation is indirectly linked to product or service innovation (Adner and Levinthal 2001; Damanpour 2020). When companies develop innovative products or services, they must substantially reconfigure their existing processes or develop new ones (Mikalef and Krogstie 2020). For instance, Fritsch and Meschede (Fritsch and Meschede 2001) have empirically demonstrated that process innovation has a positive impact on product and service innovation. More specifically, by boosting process innovation, an organization may enhance quality, introduce new products and services or include emerging technologies (Adams et al. 2006). Secondly, many manufacturing companies have begun to adopt a so-called servitization approach, in which large parts of revenues are generated from the services provided around products (Kowalkowski et al. 2017). Therefore, designing and improving the processes that underpin these services (i.e., to strengthen an organization's process innovation capability) are highly important for obtaining competitiveness and sustained performance (Trkman et al. 2015). While previous studies primarily address PIC for general business processes, it is essential to recognize the distinct nature of LSBP and their characteristics, which is the scope of the present study.

Business processes are chains of events, activities, and decisions that add value to organizations (Dumas et al. 2018). While different process types exist, they are typically categorized by their value creation perspective of core, support, and management processes. Alternatively, process types can also be classified based on other characteristics, such as their degree of repetitiveness, knowledge intensity, interdependence, and variability (Kerpedzhiev et al. 2021). Authors have also categorized process types based on their structure. First, fully structured processes are characterized by fixed rules and completely pre-defined fragments. Such processes are static with a detailed description of repetitive activities (e.g., an automated regulatory process). Secondly, unstructured processes exist when it is impossible to describe their flow in detail, yet having clearly defined goals (e.g., for investigation) (Richter-von Hagen et al. 2005). Lastly, semi-structured processes involve a combination of structured and unstructured processes. They can be further classified into two subtypes, namely (1) structured with an ad hoc exception feature as well as detailed descriptions of the process flow and defined decisions to execute one-off, ad hoc tasks or exceptions to the flow (e.g., financial back office transactions), and (2) unstructured processes with predefined fragments that are characterized by having a detailed flow description only for selected fragments (e.g., insurance claim) (Szelagowski and Lupeikiene 2020). Alternatively, Karande and Kalbande (Karande and Kalbande 2015) have classified LSBPs into four other subtypes, namely (1) sequential and simple processes (i.e., characterized by a fixed sequence), (2) complex collaborative processes (i.e., which involve complex, dynamic, and interactive aspects), (3) social environment processes (i.e., highlighting social constructs and emphasizing more the people side), and (4) feedback-based business processes (i.e., which can be modeled using systemic dynamics modelers). If business processes are knowledge-intensive and highly dynamic, the process activities cannot be completely predefined because they are dependent on decisions being made at runtime. Such process executions require more flexibility, and only loosely specified processes can support such

complex scenarios (Reichert and Weber 2012). For instance, business processes in the healthcare industry are an illustration of both structured and unstructured processes. Administrative procedures (e.g., patient registrations and discharges) are SBP examples, but also other diagnostic and treatment delivery activities (e.g., patient transfers, bookings, and lab testing) are typically structured, stable, and repetitive. In contrast, the diagnostic and therapeutic activities driven by clinical decision-making and medical case data are knowledge-intensive steps in only loosely structured or unstructured processes. Each team member's decision could be based on own experience and knowledge, or it could be the result of a group decision-making process (Di Ciccio et al. 2015).

2.2 Comparison of innovation capability frameworks

Previous studies have proposed different innovation capability frameworks to help organizations build their innovation capability using various theories, among others the resource-based view (RBV) and the dynamic capability theory (Agrawal et al. 2019; Bouwer 2016; Crossan and Apaydin 2010; Hirte and Roth 2018; Mendoza Silva 2020). For the purpose of this study, we have compared four related frameworks in Table 1 in order to determine the starting point of our work. We have started by considering two innovation capability frameworks, namely: [1] the framework of Lawson and Samson (2001), as being the most cited paper in the domain and which is based on a literature review and a case study, and [2] Mendoza-Silva (2020), as being the most recent article and covering a systematic literature review. Additionally, we

Table 1 Comparison of innovation capability frameworks

People-Process-Technology framework (Commerce 2007; Cooper et al. 1964; Goni and Van Looy 2022; Prodan et al. 2015)	Goni and Van Looy (Goni and Van Looy 2022)	Lawson and Samson (Lawson and Samson 2001)	Mendoza-Silva (Mendoza Silva 2020)	Frishammar et al. (Frishammar et al. 2012)
People pillar	Entrepreneurial capabilities	Vision and strategy	-	Strategy
	External partnerships	-	Interorganizational managerial determinants	Collaboration
	Employee skills	Creativity and idea management	-	-
Process pillar	Business process type	Organization structure and systems	Intra organization	-
	Knowledge & resource allocation	Competence base	-	-
	-	Culture and climate	-	Culture
	-	Organizational intelligence	-	-
Technology pillar	IS capability	Management of technology	-	-

have compared two process innovation capability frameworks from: [3] Frishammar et al. (2012), as being a generic process innovation capability framework with data gathered from a literature review, and [4] Goni and Van Looy (2022) for covering a process innovation capability that is specifically build for LSBP, and this also with data gathered from a systematic literature review. This means that, although we have detected multiple related capability frameworks, a thorough and empirically validated translation into an LSBP-specific context still appeared to be missing in the body of knowledge.

As an established way of scrutinizing the related frameworks' comprehensiveness and breadth of their capability coverage, we have looked at the three pillars of people, process and technology (PPT) as an initial benchmark for framework comparison. This decision has been made due to PPT's widespread acknowledgement and adoption across diverse organizational contexts. For instance, PPT has been used for understanding customer-centric processes (Chen and Popovich 2003), bridging people and software through process technology (Dumas et al. 2005), and exemplifying a Toyota product development system (Morgan and Liker 2020). The PPT roots trace back to Leavitt's model in 1964 (Cooper et al. 1964), and evolved with the Information Technology Infrastructure Library (ITIL) in the 1980s by focusing on aligning IT with business objectives (Commerce 2007). Moreover, Prodan et al. (Prodan et al. 2015) have expanded the model to include organizational culture, customer focus, innovation and management functions, thus illustrating the dynamic nature of PPT. Similarly, given PPT's intensive use and adaptability, we have also chosen for PPT as a basis for comparing related innovation frameworks, and providing a versatile reference point to evaluate their nuances.

Although the specifics or names of sub-capabilities may differ among the innovation capability frameworks in Table 1, many similarities are present (though not necessarily for LSBP and mainly based on literature reviews). For instance, as shown in Table 1, almost every capability in the related frameworks has also been covered by the LSBP-specific framework of Goni and Van Looy (Goni and Van Looy 2022). For example, the sub-capabilities of vision, strategy, interorganizational, managerial determinants, and collaboration are covered by the "People" pillar. This pillar includes entrepreneurial strategy, referring to an organization's ability to establish an innovation strategy within the overall organization strategy in order to balance between strategic vision and operational excellence. Additionally, it addresses strategies to effectively convey their strategic message to employees. The "Process" pillar covers areas for organization structure, intelligence, culture and competence base, whereas the management of technology is included in the "Technology" pillar.

In sum, despite earlier literature emphasizing the need of developing PIC in LSBP (vom Brocke et al. 2015), we have discovered only one generic PIC framework (Frishammar et al. 2012), but without distinguishing between SBP and LSBP. On the other hand, Goni and Van Looy (2022) comply to all PPT pillars when deriving their LSBP-specific PIC framework from the body of knowledge (albeit without empirical data) (Goni and Van Looy 2022). Recognizing this gap, we have decided to adopt the framework from Goni and Van Looy (2022) as a starting point. In the remainder, we will apply the Delphi method to empirically validate and extend the knowledge for PIC in LSBP. Our objective is not only to refine and enhance the related capability

areas. This study will also utilize these capability areas as the basis for transitioning from a conceptual to a practical overview of assessable or measurable PIC-LSBP capabilities. Our main contribution thus lies in first delineating the differences between LSBP and SBP for each capability, while also refining the sub-capabilities and especially providing reflection questions that organizations can use to brainstorm about their current PIC in LSBP.

3 Methodology

3.1 Delphi motivation

The Delphi method is a research method to obtain consensus of opinions about a particular problem or topic throughout multiple iterations of anonymous questionnaires, and interspersed with feedback (Dalkey and Helmer 1963; Landeta 2006; Paré et al. 2013). Delphi studies have been widely used for research justification and theory development in a variety of research disciplines (Hsu and Sandford 2007), including the field of business process management (Becker et al. 2015; Bruin and Rosemann 2007a; Kerpedzhiev et al. 2021; König et al. 2019; Schmiedel et al. 2013; Van Looy et al. 2017).

While different types of Delphi studies exist (Van Looy et al. 2017), we followed the modified Delphi approach. This method is similar to a classic Delphi in terms of its protocol, including the number of rounds and the intention of reaching consensus. However, the modified approach differs depending on the nature of the project (Richards and Curran 2002). The starting procedure for selected items can be different, such as using literature reviews to provide basic information concerning the topic (Custer et al. 1999). While the first round in a classical Delphi begins with an open qualitative round to let panelists record their responses from scratch, panelists in the modified Delphi are given pre-selected items collected from various sources and are asked to examine their responses (Hasson and Keeney 2011). Therefore, in the first phase of our study, we provided the experts with a list of definitions and initial process innovation capability areas, as derived from the systematic literature study of (Goni and Van Looy 2022). Additionally, we asked experts to provide their opinion about the main barriers and differences between innovating structured and less-structured business processes per capability area, but without showing the framework.

Another reason to motivate our Delphi choice is because those studies generally result in a higher quantity and quality of ideas than other group decision-making methods (Van De Ven, Delbecq, & L, 1974). Expert anonymity allows for deeper richness of the data and minimizes dominant personalities in face-to-face interactions (Murphy et al. 1998). It helps avoid direct confrontation among the experts, and eliminates geographical boundaries and associated travels (Okoli and Pawlowski 2004). For ensuring anonymity, we allocated an anonymous and unique ID per expert (Okoli and Pawlowski 2004).

3.2 Expert selection

Experts from both academia and practice were selected to ensure a balance of views from both theoretical and practical perspectives (de Bruin and Rosemann 2007a), as demonstrated in Appendix A. The academics had a PhD and minimal two academic publications that combined the topics of innovation and BPM in academic outlets with an impact factor in the Social Science Citation Index (SSCI). Still, we decided to keep one academic expert without SSCI publications to guarantee sufficient experience in one continent, but who included more than a hundred academic publications and over one thousand Google Scholar citations at hand. On the other hand, the practitioners were consultants and senior managers in innovation who applied BPM in medium to large-sized organizations, both profit and non-profit sectors, and with a minimum experience of seven years. We aimed for at least one expert per continent to introduce diverse geographical backgrounds and minimize bias (de Bruin & Roseman, 2007). We used LinkedIn to contact the experts and asked them to nominate other experts if they were unable to participate themselves.

We followed the selection procedure of Okoli and Pawlowski (Okoli and Pawlowski 2004), in which 18 to 22 experts were recommended. Preparatory work for the expert selection and evaluation started by the end of August 2021, and the initial phase was launched in October 2021 when sending out forty-four invitations across five continents. Nineteen invitations were sent to practitioners and twenty-five to scholars, of which nine experts were referrals. The response rate for the first and second rounds was 77% and 90%, respectively. Figure 1 summarizes the final expert composition per round.

3.3 Data collection and analysis

Adapted from Schmiedel et al. (2013), Table 2 synthesizes each Delphi round. This overview covers three rounds: (1) brainstorming about the translations and important factors derived from the selected work in [Comparison of innovation capability frameworks](#), namely by Goni and Van Looy (2022), (2) narrowing down the original list to the most important ones by ranking, and (3) narrowing down until consensus was reached and deriving the intended reflective questions for a practical brainstorming.

The Delphi study took nine months in total. We began with preparations for the panel creation, followed by three Delphi rounds. We started by selecting experts based on the minimum requirements and by evaluating their expertise. In each round, the experts were asked for any suggestions or comments for changes (i.e., additions, deletions, expansions, or further comments) to the framework elements. We also intended to quantitatively evaluate the framework by assessing the experts' overall satisfaction (Kerpedzhiev et al. 2021). Moreover, in the second and third rounds, we also asked for the respondents' overall satisfaction per capability. The data was analyzed using MS Excel and SPSS. Per round, the experts had one month to provide feedback and they were free to use e-mails or Zoom meetings, whichever was most convenient to them. Given that a Delphi study requires more time from respondents than a traditional survey (due to the multiple rounds), one goal was to ensure that a single questionnaire would not take more than 45–60 min to complete.

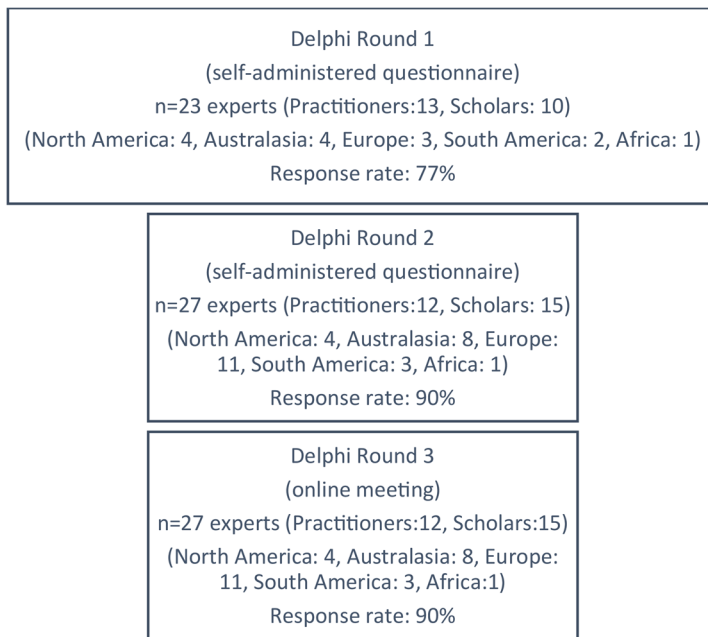


Fig. 1 Expert composition by region and Delphi round

In the first round, we sent a questionnaire in October 2021 with open-ended questions to generate ideas for translating the differences between SBPs and LSBPs (Delphi Round 1_Main differences and Barriers). We divided the questionnaire into two sections. First, we explained the study's purpose and the definitions to be used. Secondly, we requested the experts to provide their expert opinion on the main differences and barriers when innovating SBPs and LSBPs. The expert then filled in their opinions on each of the framework elements using the initial capability framework from the literature([Comparison of innovation capability frameworks](#)). The output from this phase was a summary of the differences and barriers between innovating in SBP versus LSBP, and an overview of extra capabilities that were not in the initial list.

In the second round, the questionnaire was sent at the end of December 2021 and compiled by mid-February 2022. Our objective was to validate the refined framework. The first part of the questionnaire was a consolidation of answers, critics, and suggestions from round 1, which were now re-classified into pillars. For the second part, we presented a summary of the LSBP-specific keywords and asked the experts to rate the extent to which they agreed with each difference and barrier using a 7-point Likert scale (1 = totally disagree; 7 = totally agree) as well as to rank the importance of each capability (Delphi Round 2_Rating and Ranking). As a result, we analyzed the outcome, we extended and validated the capabilities, and prepared an initial version of a practical instrument in response to experts' suggestions. The practical instrument was intended as a brainstorming in the form of reflective questions based on each capability in the framework.

Table 2 An overview of Delphi rounds

Round	Input	Output
Preparatory phase (Panel Creation) (Aug-Sept 2021)	• Start the expert selection • Do a competence assessment of the experts	• Confirmed list of experts
1st round – Brainstorming about translations (Oct-Nov 2021)	• Explain the study's purpose and the definitions that will be used • Show each of the elements based on the initial capability framework from the literature • Ask panelists to provide their expert opinion per capability	• List of differences and barriers between innovating in SBP versus LSBP • List of capabilities that are not in the initial list
2nd round - Narrowing down by ranking (Dec 2021-Feb 2022)	• Summarize the differences and barriers from the first round, reclassify the capabilities, and request a rating • Ask panelists to rate the importance of each capability based on the results from round 1 and request for suggested changes	• Agreed formulations per capability (5-point Likert scale), without showing the pillars • Agreed importance of each capabilities • Group of capabilities across PPT pillars
3rd round and subsequent rounds, until consensus was reached) - Narrowing down the framework and brainstorming instrument (March-June 2022)	• Verify the answers, comments and suggestions from the previous round • Present the instrument procedure along with the framework for brainstorming • Verify consensus or a next iteration	• Reconfiguration of capabilities • Ranked overall satisfaction of the framework • List of reflective questions in a brainstorming instrument
Wrap-up phase – Evaluation (June-July 2022)	• Determine the final capabilities • Summarize the study	• Validated conceptual framework • Validated practical guidelines

In the third and final round of March 2022, we consolidated all answers, critics and suggestions from round 2, and reconfigured the final capabilities in a conceptual framework along with practical guidelines for the intended brainstorming. We divided the document into two parts and asked the experts to review the practical instrument procedure (part 1) and the reflective questions (part 2) to translate the framework into a more practical overview (Delphi Round 3_Instrument). These final interviews were concluded on June 21, 2022.

3.4 Research validity and reliability

A measurement procedure is generally seen as valid when it consistently produces the research results (Golafshani 2003; Kirk 1986; Seale 1999). Therefore, per round, the experts' opinions and suggestions were consolidated by the researchers, and shared in a next round until predefined criteria were met (Paré et al. 2013). More specifically, consensus criteria were developed to indicate when a satisfactory level of consensus had been reached and to determine the number of rounds (Mullen 2003). We continued with extra rounds until the conditions were met. In this study, consensus was reached after three Delphi rounds. We applied ratings on a 7-point Likert scale

(1 = extremely unimportant; 7 = extremely important). These consensus criteria were as follows (Diamond et al. 2014; Hsu and Sandford 2007; Van Looy et al. 2017):

- 50% of the experts must agree on the two most extreme scores (i.e. either 1–2 for consensus to exclude a criterion or 6–7 for consensus to include a criterion), and.
- Seventy-five must agree on the three most extreme scores (i.e. either 1–2–3 or 5–6–7), and.
- The interquartile range (IQR, i.e., the distance between the 75th and 25th percentile, or the range of the middle 50% of responses for a criterion) must be 3 or less, and.
- No opposite extreme score is given by any expert (i.e., either no 7 or no 1 for a criterion).

In addition to the consensus conditions, the study would stop if the majority of experts stated they were no longer willing to iterate. Hasson et al. (Hasson and Keeney 2011) and Mullen (Mullen 2003) recommended to have a response rate of at least 70% of the initial panel size to preserve rigor and validity. We concluded with the third round, as consensus had been reached and the experts appeared to be no longer willing to continue. There was no further feedback for the framework during the third round, only requests from the majority of experts to empirically test the framework using the established instrument (e.g., in case studies as a follow-up study).

Furthermore, we asked the experts to assess their overall satisfaction score of the framework, for which we obtained consistent results from the second to the third round (i.e., from 7.52 to 8.36 out of 10) as shown in [Round 3: practical overview with assessment questions for reflecting on innovating LSBP](#), with no further significant additional feedback. Our Delphi study with three rounds was distributed to experts in the predefined criteria, covering both scholars and practitioners from five continents. The use of subject matter experts (i.e., with knowledge of and interest in the topic) promoted content validity, while the use of multiple rounds increased concurrent validity (Hasson and Keeney 2011).

Several authors argued that the Delphi method could enhance reliability (Alarabiat and Ramos 2019; Hasson and Keeney 2011; Lilja et al. 2011). The reliability of the method was assessed based on three items: the size of experts, the selection of experts, and the conducting of the process (Lilja et al. 2011). While some debate exists about whether the number of experts is a reliable indicator (Delbecq et al. 1975; Ziglio and Adler 1996), we had 27 experts worldwide (thus more than the minimum of 15). However, a major consideration was the experts' knowledge and experience in the topic. Therefore, the selection of experts (i.e., with knowledge and interest in the topic) was crucial (Alarabiat and Ramos 2019; Goodman 1987; Hasson and Keeney 2011; Lilja et al. 2011). In this study, experts were carefully selected from both academics and practitioners in the area of the investigated topic. We maintained a consistent number and composition of experts between academics and practitioners per round, demonstrating their commitment to the research. Thirdly, experts had the possibility to iteratively evaluate their responses and to comment on other answers, allowing for a relatively high level of reliability (Hasson and Keeney 2011; Landeta 2006). Experts' anonymity was also maintained to prevent group bias and the occur-

rence of group think scenarios, enhancing reliability as well (Alarabiat and Ramos 2019).

4 Results

Subsequently, we present the results per Delphi round.

4.1 Round 1: main differences and barriers for innovating in LSBP versus SBP

The first Delphi round yielded five main differences (Table 3) and barriers (Table 4) between innovating structured (SBP) and less-structured business processes (LSBP), as the first novel insight in this study.

We summarized the experts' inputs and categorized them into five main barriers, as shown in Table 4.

Table 3 Main differences between innovating in LSBP and LSBP

Difference	LSBP	SBP	Number of experts
Flexibility	LSBP is less formal. The procedural flow (e.g., rules and fragmented order or sequence of steps) is more varied and agile. Process execution can be altered (e.g., by making it less varied or less repeatable). LSBP tends to be faster and easier to identify because the flow of activities is less standardized with fewer systematic restrictions or manual procedures, making it easier to change.	SBP has a strong institutionalization with a well-defined workflow and clear measurement system.	11
Uncertainty	LSBP carries a high level of uncertainty and unavailability of baselines on how the process should be executed for its best performance. Since parameters vary from case to case, it is difficult to determine if the process is reaching its best performance.	SBP has a more standardized state with clear paths and rules that can be used as a benchmark for comparison.	9
Empowerment	The individuals involved in LSBP are using their own judgement to make decisions about how work is processed, rather than relying completely on pre-determined rules and flows.	In SBP, individuals adhere to the defined rules and flows.	2
Collaboration	LSBP is more about connecting people, fostering creativity, and enabling the process to adapt constantly to new contexts. LSBP is linked to informal social interactions, where tacit knowledge, values, and mental models can be shared.	SBP focuses more on systems of business activities and explicit knowledge.	3
Software support	LSBP has a more dynamic and organic flow. The software systems available to help automate LSBP are limited so far. LSBP therefore requires more creative solutions to understand alternative technological solutions for improving and innovating the process, such as machine learning and artificial intelligence.	SBP is typically well equipped with software-based monitoring systems that enable them to keep track of process steps, challenges and opportunities, resource allocation, and relationships.	6

Table 4 Main barriers when innovating in LSBP and SBP

Barrier	LSBP	SBP	Number of experts
Control	LSBP faces a challenge to implement the required process changes due to a lack of control.	SBP typically has a well-defined procedure and control in place.	10
Outcomes	It is typically more difficult to determine the impact of innovation outcomes in LSBP. Possible explanations are a lack of vision for the TO-BE process in order to accurately capture the related business problem or difficulties to link objectives to consumer expectations.	SBP has more clarity with respect to its parameters and the innovation path of organizations.	6
Technology	Current process modeling notations are still less suited to design LSBP due to their failing to identify those parts of the process that could be standardized and failing to recognize the potentials of new technologies such as machine learning and AI (e.g., to automate or make recommendations for human decisions).	SBP profits from more well-known software in the market.	5
People	Change in LSBP requires more alignment of culture, structure, techniques and technology because of a high dependency on knowledge workers' expertise to perform the tasks. Teams that have been able to operate in a non-structured way should understand and accept the need for a process change.	SBP has pre-defined rules that can be followed more easily by employees.	9
Time and Budget	Time and budget are more critical in LSPB because innovation necessitates a thorough understanding of the process while LSPB has uncertain characteristics. A significant amount of time and effort will be needed to comprehend LSPB.	SBP has a clear flow, requiring less time and effort to understand the process to innovate.	2

We asked experts to provide differences between innovating in LSBP and SBP and also to provide extensions to the literature-based framework of Goni and Van Looy (Goni and Van Looy 2022). While the results (shown in Fig. 2) suggested to maintain the main capabilities identified in this literature-based framework, the procedure for refining the sub-capabilities involved several stages, including organizing relevant capabilities, eliminating, and integrating sub-capabilities to enhance the framework's applicability. The intermediate framework resulting from this round included 27 sub-capabilities across three main pillars (i.e., People, Process, Technology pillars). It consisted of six main capability areas: [1] Employee skills, [2] Network of external relationship, [3] Entrepreneurial, [4] Business process type, [5] Knowledge resource and allocation, and [6] Information systems (IS). Descriptions per capability are available online (Delphi Round 1_Main Differences and Barriers). The experts had no objections to the proposed capabilities.

At the end of round 1, the experts were asked for missing or under-emphasized elements, resulting in six suggestions that we incorporated into the proposed capabilities of Fig. 2 as follows (Table 5).

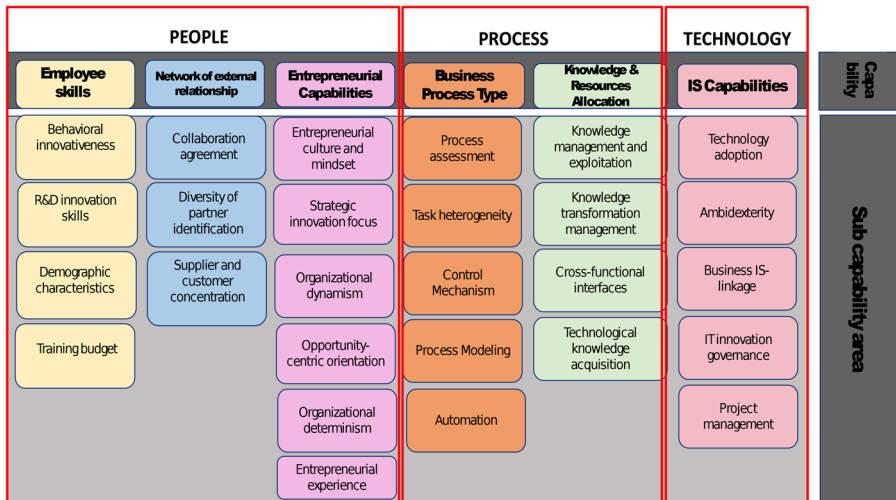


Fig. 2 Intermediate core elements of PIC in LSBP after round 1, and partially extending the selected framework

Table 5 Proposed additional capabilities

Other capabilities	Included in
Change management includes legislation, governance, rules and norms, an approach to process innovation (e.g., reactive-proactive, enhancement vs. design, etc.). A stronger focus is needed on technology, rewards and motivation, and empowerment.	Category 2-Knowledge management and Exploitation
Budget, and particularly Capital Expenditure (CAPEX) and intellectual property (patents)	Category 6- Technology adoption now includes patents as one of the example
Alignment of culture-structure-techniques-technology	Category 2-Knowledge management and Exploitation
Soft skills regarding creativity, social intelligence, innovativeness and complex problem solving to support innovation, decision making, and to respond to wicked problems	Category 3-Behavioral & Innovation skills
Strategy in general as a separate capability to emphasize an organization's vision	Category 5-Strategic innovation focus
Management maturity levels, including process governance (e.g., how roles and responsibilities are structured), performance management (e.g., the adoption of KPI's, goals, targets and control items) and continuous improvement (e.g., existence of methods, tools and problem-solving practices on a daily basis).	Category 1_Control mechanism and Category 6_IT innovation governance and Business IS linkage

4.2 Round 2: rating and ranking

In the second Delphi round, the differences and barriers from the first round were rated. The observed differences per sub-capability from the first round were provided as input for the second round and are available online (Delphi Round 2_Rating and Ranking). The experts were also asked to rank the LSBP innovation capability sub-

Table 6 Ratings of the main differences

Criteria/Difference	Flexibility	Uncertainty	Empowerment	Collaboration	Software support
50% of the experts agree on the two most extreme scores (i.e., 6–7),	62%	81%	50%	50%	50%
75% of the experts agree on the three most extreme scores (i.e., 5–6–7),	88%	96%	85%	81%	73%
The interquartile range must be 3 or less	1	1	2	2	2

Table 7 Ratings of the main barriers

Barriers/Criteria	Control	Outcomes	Technology	People	Time & budget
50% of the experts agree on the two most extreme scores (i.e., 6–7),	56%	52%	33%	67%	52%
75% of the experts agree on the three most extreme scores (i.e., 5–6–7),	85%	63%	63%	85%	63%
The interquartile range must be 3 or less	2	2	2	2	2

areas in order of importance (i.e., per main area). For instance, rank 1 represented the highest importance with respect to the other sub-areas within the same category, rank 2 represented the second-highest importance for innovating in LSBP, etc.

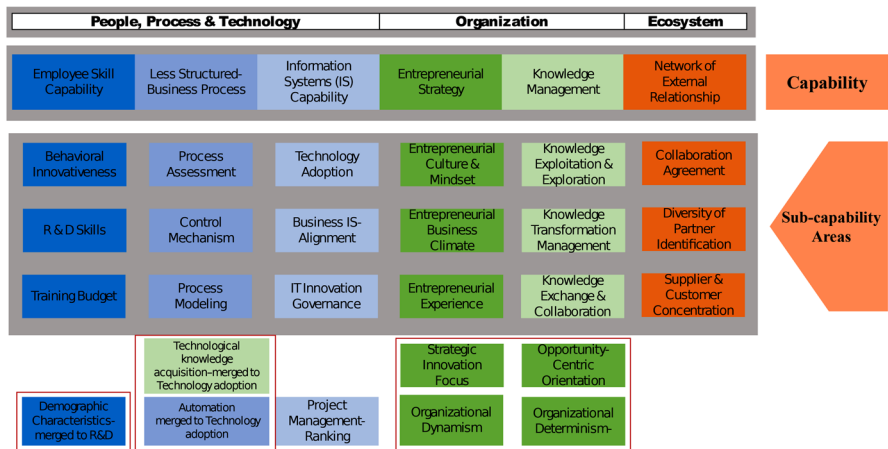
As shown in Table 6, all differences (i.e., flexibility, uncertainty, empowerment, collaboration and software support) met the predefined consensus criteria of [Research validity and reliability](#). We summarize the ratings for barriers (control, outcome, technology, people, time and budget) in Table 7. Only the results for the technology barrier did not meet the minimum of 50% of experts agreeing on the two most extreme scores (i.e., 6–7), while only the barriers related to control and people met the 75% agreeing on the three most extreme scores (i.e., 5–6–7).

Further on, none of the experts had objections against the proposed capabilities. The capability related to business process type received the most input with 22 expert comments, while the network of external relationships and the information system capability received the fewest. In addition, we calculated the interquartile range (IQR) based on the consensus criteria, yielding the result shown in Table 8. The IQR was less than 3, and no expert provided a score at the opposite end of the spectrum (i.e., 1). All capabilities thus met the consensus requirements.

At the end of round 2, we asked experts to review their input from the first round and to provide additional feedback per sub-capability, if any. The majority of feedback concerned terminology, particularly regarding the labeling or overlap of sub-capabilities, but no substantial concerns related to the capabilities. Figure 3 illustrates the version of our intended framework after round 2. According to six experts, seven sub-areas seemed to have some overlap: [1] Demographic characteristics, [2] Automation, [3] Technological knowledge acquisition, [4] Strategic innovation focus, [5]

Table 8 Number of experts' comments and IQR per capability

Capability	Experts ID	Q1	Q2	Q3	IQR
Business process type	1–4, 6–23 (22 experts)	8	8.5	9	1
Knowledge management and resource allocation	3,4,6,8,10–23 (17 experts)	7	9	9	2
Employee skills	4,6,9,11, 12,14–23 (15 experts)	8	9	9	1
Network of external relationships	4,6,8,10–12,15–17,19–23 (14 experts)	7	9	9.4	2.4
Entrepreneurial strategy	1,3,4–6,8–12, 14,17–19,21–23 (17 experts)	8	9	9	1
Information systems	1–7,9–11, 14,16,18,19 (14 experts)	8	8.5	9.8	1.8

**Fig. 3** Intermediate core elements of PIC in LSBP from the expert panel feedback after round 2

Organizational dynamism, [6] Opportunity-centric orientation, and [7] Organizational determinism. Therefore, in our iterative approach, we first relabeled and re-configured the sub-areas to make them easier to understand and better fit into their definitions to obtain the final expert feedback and consensus (i.e., indicated as red boxes in Fig. 3). We then extended the framework by removing duplicates, re-labeling the elements, and integrating them: [1] demographic characteristics were merged into R&D, [2] automation and [3] technological acquisition were merged into technology adoption, and entrepreneurial business climate was comprised of [4] strategic innovation focus, [5] organizational dynamism, [6] opportunity-centric orientation, and [7] organizational determinism. The order and number of each capability was determined by its ranking from the second round. Project management was removed due to a low-ranking result.

4.3 Round 3: practical overview with assessment questions for reflecting on innovating LSBP

Throughout the study, the overall quality-related satisfaction increased. We asked experts to judge the framework's overall score, which increased from 7.52 out of

10 in round 2 to 8.36 out of 10 in round 3. In the third round, experts also provided feedback related to the intended brainstorming, namely on the overview of the instrument procedure and the reflective questions derived from the validated framework elements. These reflective questions were a translation of the conceptual framework into a practical overview, acting as a solution to guide brainstorming and assist organizations in assessing their current innovations capabilities within an LSBP context.

As a continuation of the second round, the experts asked for a demonstration of how the framework would be implemented in organizations. Hence, we prepared a practical instrument based on the framework elements. Each capability was incorporated into the reflective questions, for an organization to assess its current capabilities. To construct these reflective questions, we relied on the initial SLR definitions and expert inputs from prior rounds as a base. We also designed a step-by-step procedure for illustrating the instrument's use (Fig. 4).

We offered a total of 55 reflective questions, each with 3 to 4 questions per sub-capability that organizations could use. For instance, Table 9 illustrates the reflective questions related to one capability of knowledge management. The complete brainstorming is available online (Delphi Round 3_Instrument).

5 Discussion

Our primary contribution is a dedicated framework for PIC in LSBP ([Interpretation of the main findings](#)). We also compare this resulting framework to existing capability frameworks ([Theoretical contributions](#)) for better indicating the need for new capability areas to address the identified barriers and differences in LSBP innovation. Next, we discuss the practical contributions ([Practical contributions](#)) as well as the research limitations and avenues ([Research limitations and future research](#)).

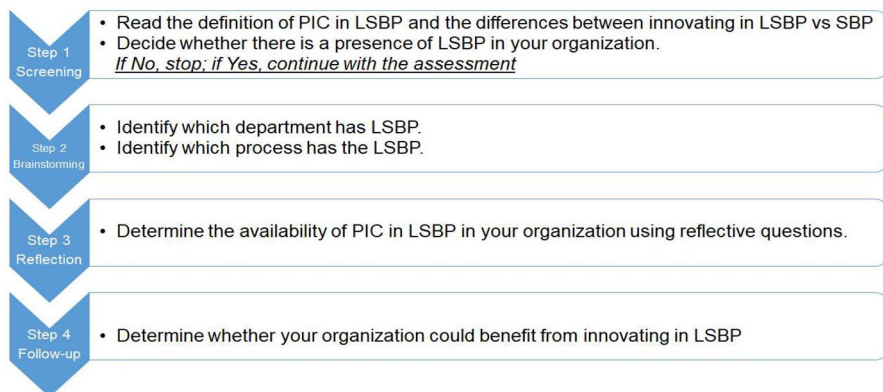


Fig. 4 Step-by-step procedure for brainstorming about PIC in LSBP

Table 9 Example of reflective questions for the main capability of “Knowledge Management”

Knowledge management (knowledge across the entire organization)		
Knowledge creation and sensing (= an organization's ability to adopt, refine, synthesize existing knowledge, and search for new knowledge beyond the organization's existing expertise)	LSBP innovation is characterized by more undocumented and tacit knowledge and silos. Organizations first gain access to different ideas and knowledge through collaboration, before leveraging existing expertise and exploring new opportunities. This knowledge assists in the creation of new process innovation. On the other hand, innovation in SBP is more defined and well documented, being easily accessible and manageable because of well-defined processes that allow for ideas through internal development.	1. In your organization, what possibilities exist to facilitate cross-functional interfaces? 2. In your organization, what possibilities exist to exchange knowledge mostly with business partners rather than internally? 3. In your organization, what possibilities exist to involve employees of other departments to help solve a problem faster and without time-consuming procedures [e.g., collaboration projects with a different skill set, dedicated project management software)?
Knowledge transformation (= an organization's ability to combine and converse existing and new acquired information and knowledge into valuable novel ideas)	LSBP requires more effort to transform knowledge to new ideas. On the other hand, SBP has a lower need for knowledge transformation since the existing innovation solutions often propose standard patterns and ready-made software can be more easily adapted to process needs	1. In your organization, what possibilities exist to transform knowledge into valuable ideas for innovating LSBP (e.g., workshop, meeting, modeling)? 2. In your organization, what possibilities exist to reconfigure explicit knowledge (e.g., using document management systems that offer features such as storing/uploading of files, organizing documents, searching and retrieving based on indexing techniques, advanced searching mechanisms and access from any internet-connected workstation)? 3. In your organization, what possibilities exist to transform knowledge into new ideas in LSBP innovation projects (e.g., using different terminology, with unclear project goals, difficulties in finding employees the exact knowledge)?
Knowledge sharing and dissemination (= an organization's ability to exchange knowledge across the entire organization)	LSBP needs more intense communication, centered around individual expertise, with less demarcation that allows less bureaucracy to reduce time-consuming procedures. Knowledge workers may choose to involve other people outside their own department to help solve a problem faster, and this with less time-consuming procedures. On the other hand, SBP has more knowledge distribution within the organization for which the communication mechanisms are already in place, well documented and easier to access, but usually by following a more rigid procedures that take more time.	1. In your organization, what possibilities exist to facilitate cross-functional interfaces? 2. In your organization, what possibilities exist to exchange knowledge mostly with business partners rather than internally? 3. In your organization, what possibilities exist to involve employees of other departments to help solve a problem faster and without time-consuming procedures [e.g., collaboration projects with a different skill set, dedicated project management software)?

5.1 Interpretation of the main findings

Our study has gradually built and tested a conceptual framework for PIC in LSBP, as shown in Fig. 5. The framework consists of three main pillars, namely first the people-process-technology (PPT) pillar, then the organizational pillar and lastly the ecosystem pillar. In total, it includes 18 sub-capabilities across six main capability areas: [1] Employee skills, [2] Less-structured business processes, [3] Information systems (IS), [4] Entrepreneurial strategy, [5] Knowledge management, and [6] Network of external relationship.

First, the “PPT pillar” consists of three main capabilities and three sub-areas each, as illustrated in Table 10. The main capability within PPT, “employee skills” [1], emphasizes the importance of employee-specific R&D skills to develop new products or services with LSBP characteristics. It consists of three capability sub-areas. First, employee’s behavioral innovativeness [1.1] highlights the need for a more dynamic work profile in LSBP to identify potential solutions for a variety of scenarios, and for determining the most appropriate technological fit. Secondly, R&D skills [1.2] focuses on the importance of having employees with past professional experience in R&D related to data science, machine learning, and artificial intelligence (AI), and with a broad knowledge in new technologies. These employees can identify potential solutions for a variety of scenarios and for determining the most appropriate technological fit. Lastly, the training budget [1.3] emphasizes the importance of a sufficiently large budget to cover training costs for innovation in LSBP. LSBP necessitates diverse training, often dedicated training per LSBP and with less reuse, thus requiring higher training costs. The second main capability within PPT, “LSBP characteristics” [2], highlights the characteristics of a process assessment [2.1] in LSBP, which is more difficult due to a lack of benchmarking, higher level of customization, and lack of structure. Hence, an organization must have a control mechanism [2.2]

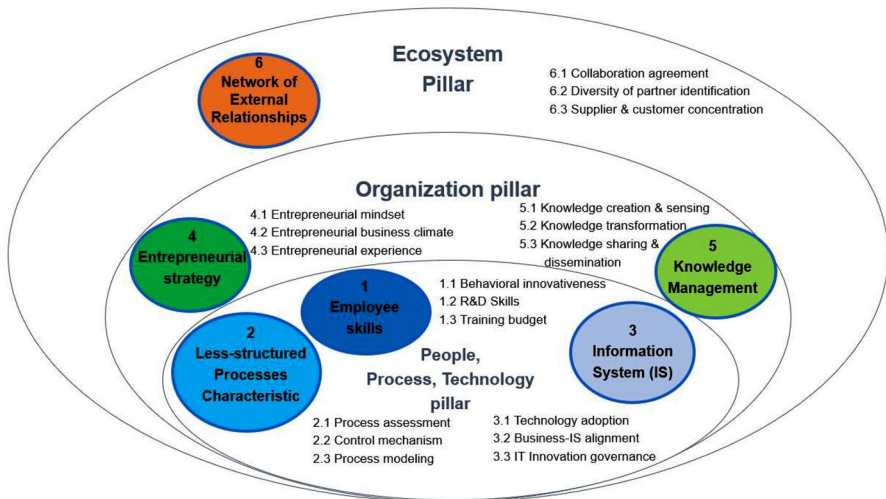


Fig. 5 Empirically validated framework of PIC in LSBP

Table 10 Overview of the PPT-related capability areas

Main capability	Sub-capability	Definition
[1] Employee skills: employee-related attributes (i.e., knowledge, skills, and behaviors) that need to be adapted in order to improve communication and foster innovation	1.1 Behavioral innovativeness	The presence of employees with the ability to identify, understand, build and foster innovation
	1.2 R&D skills	The presence of employees with past professional experience in research and development, and skills to develop new products or processes
	1.3 Training budget	The presence of a sufficient and dedicated budget to cover training costs related to the innovation of less-structured business processes
[2] LSBP characteristics: business processes characterized by unpredictable situations that are flexible and more complex, involving ad hoc tasks with limited availability of detailed flows, dynamic collaborations and people-centric	2.1 Process assessment	An organization's ability to evaluate a less-structured process by comparing it to best-in-class processes or by using established quality standards
	2.2 Control mechanism	An organization's ability to regulate less-structured activities by establishing formalized procedures and business rules
	2.3 Process modelling	An organization's ability to facilitate the flow of a less-structured process by means of an abstract representation or visualization
[3] IS-related capabilities: an organization's ability for facilitating several ways in which IT could support process innovation and reengineering, such as for facilitating project management or regarding IS resources and competencies affecting organizational performance	3.1 Technology adoption	An organization's ability to acquire and use a new technology, machine, robot or computer-aided design, and to anticipate future technological necessities
	3.2 Business IS-alignment	An organization's ability to solve business problems by integrating or aligning IT with business needs and the organization's objectives and overall strategies
	3.3 IT innovation governance	An organization's mechanisms to assure an effective and efficient assignment of decision-making authority and to manage risks related to an innovation project, enabling an organization to achieve its goals

to regulate less-structured activities by establishing formalized procedures and business rules. LSBP has a much looser control mechanism and more measurements to aggregated outcomes. It is more dynamic, allowing individual employees to develop their rules and ways of working to allow for unstructured local decisions with more autonomy for employees. Therefore, organizations must have an ability in process modelling [2.3] to facilitate the visual flow of LSBP because of a higher percentage of exceptional cases. The third main capability, "IS-related capabilities" [3], emphasizes an organization's ability to anticipate its future technological needs through technology adoption [3.1] and business IS-alignment [3.2] by aligning IT not only with business needs, but also with the organization's objectives and overall strategies. LSBP innovation projects tend to have more specific requirements for IT innovation governance [3.3] (e.g., how roles and responsibilities are structured) that need to be translated into the business. Consequently, they require more attention to manage risks, and elevated involvement of leadership to ensure that an organization's strategic objectives are less exposed by IT failures.

Secondly, the "organization pillar" comprises two main capability areas, each with three sub-areas (Table 11). The first main organizational capability, "entrepreneurial strategy" [4], focuses on creating a more dynamic environment that helps organiza-

Table 11 Overview of the organization-related capability areas

Main capability	Sub-capability	Definition
[4] Entrepreneurial strategy: an organization's ability to explore, integrate, and exploit untapped business opportunities	4.1 Entrepreneurial mindset	A corporate culture that creates a value system for encouraging employees to continuously innovate
	4.2 Entrepreneurial business climate	An organization's ability to establish an innovation strategy within the overall organization strategy to achieve a balance between strategic vision and operational excellence and to become more dynamic in order to compete in the market or to take advantage of a business opportunity
	4.3 Entrepreneurial experience	An organization's past activities to sell or discontinue a business, innovate a business process, invest in equipment or software, invest in R&D, or acquire patents)
[5] Knowledge management: an organization's ability for organizing, creating, using and sharing collective knowledge	5.1 Knowledge creation and sensing	An organization's ability to adopt, refine, synthesize existing knowledge, and search for new knowledge beyond the organization's existing expertise
	5.2 Knowledge transformation	An organization's ability to combine and converse existing and newly acquired information and knowledge into valuable novel ideas
	5.2 Knowledge sharing & dissemination	An organization's ability to exchange knowledge across the entire organization

tions in competing and seizing business opportunities. It consists of three sub-areas. First, an entrepreneurial mindset [4.1] indicates more entrepreneurial values in LSBP because there are less pre-defined steps to be followed and by heavily relying on individual perceptions resulting in more variation by function or geography. Therefore, it requires an entrepreneurial business climate [4.2] as an organization ability to establish an innovation strategy within the overall organization strategy. It helps organizations in achieving a balance between their strategic vision and operational excellence and to become more dynamic to take advantage of a business innovation. Thirdly, entrepreneurial experience [4.3] highlights that organizations can benefit from past entrepreneurial activities when launching a new innovation and focuses more on process innovation rather than product innovation. The second main capability, "knowledge management" [5] focuses on the organizational ability to create, transform and share knowledge. LSBP innovation is characterized by more undocumented and tacit knowledge and silos. The first sub-capability, knowledge creation and sensing [5.1], emphasizes the importance of searching for new knowledge and overcoming silos. Organizations gain access to different ideas and knowledge through collaboration, before leveraging existing expertise and exploring new opportunities. Organizations must have an ability to transform knowledge into new ideas within LSBP innovation projects [5.2]. Hence, LSBP requires more intense communication to overcome silo thinking with less demarcation to allow for less bureaucracy and reduce time-consuming procedures as a third sub-areas for knowledge sharing and dissemination [5.3].

Lastly, the "ecosystem pillar" relates to a network of external relationships, which has three sub-areas as indicated in Table 12. The main capability of "network of external relationship" [6] focuses on the way to innovate without owning all capabilities but by benefiting from external collaborations instead. It consists of three sub-areas. First, the area of collaboration agreement [6.1] emphasizes the ability to

Table 12 Overview of the ecosystem-related capability areas

Main capability	Sub-capability	Definition
[6] Network of external relationship: external partnerships when an organization innovates without owning all required capabilities internally but having access to external knowledge through collaboration with external actors, relationships and resources	6.1 Collaboration agreement	An organization's ability to arrange collaboration agreements with external partners
	6.2 Diversity of partner identification	An organization's ability to screen potentially different partners in order to increase an organization's expertise by sharing resources and capabilities, and to create mutual benefits)
	6.3 Supplier and customer concentration	The total number of suppliers and customers, and the degree to which supplier/customer information affects an organization's innovation investment decisions

collaborate with external partners. LSBP innovation typically requires a variety of solutions, and external partners in both vertical or horizontal alliances can help accelerate LSBP innovation with newer technologies and capabilities that may still be unavailable within an organization. Secondly, organizations need to have the ability to identify diverse partners [6.2] to help increase expertise by sharing resources and creating mutual benefits among various partners. Lastly, a supplier and customer concentration [6.3] implies that LSBP relies more on individual decisions in partner identification, resulting in less standardization and a larger number of suppliers as the third capability sub-area. Consequently, LSBP decisions can be executed locally or per function, with less bureaucracy and being less time-consuming to accommodate different suppliers' or customers' requirements.

5.2 Theoretical contributions

Throughout the different Delphi rounds, we have iteratively refined and validated the PIC-LSBP capability elements in a conceptual framework related to a still under-investigated subject that requires more scientific attention given today's dynamic market conditions (Rosemann 2014; vom Brocke et al. 2015). In response, our findings advance the process management and innovation management disciplines by revealing the specific PIC-LSBP points of attention. We have gradually extended each finding together with an international panel of Delphi experts, while also extending the four capability frameworks of Sect. 2.2.

For determining the novelty of our identified capabilities, we have applied an interpretative approach in which we consider all capability area descriptions to stipulate whether the identified capabilities are covered by existing frameworks. Overall, we have empirically validated and refined the sub-capabilities in the literature-based framework of Goni and Van Looy (2022). First, the PPT pillar emphasizes the importance of LSBP-specific characteristics and employee skills for innovation behavior. Our research did not explicitly identify culture as one of the capabilities, but it encompasses five others (i.e., behavioral innovativeness, control mechanism, IT innovation governance, business IS linkage, and entrepreneurial mindset). In addition, Lawson and Samson (2001) and Mendoza-Silva (2020) focus on the utilization of internal technology to facilitate innovation behavior, whereas Frishammar et al. (2012) focus

on the creation and development of new process technologies. Our study thus anticipates future technology necessities.

Secondly, the organization pillar includes capability areas such as the strategic alignment capability from Frishammar et al. (2012), and the corporate vision strategy from Mendoza-Silva (2020) and Lawson and Samson (2001). Nevertheless, our focus is more on the entrepreneurial business climate to balance a strategic vision with operational excellence. In our research, the entrepreneurial mindset emphasizes a corporate culture, whereas the other frameworks emphasize more a managerial commitment or attitude (Frishammar et al. 2012), and a management style in relation to the corporate culture (Mendoza-Silva, 2020). The corporate culture shapes a value system that encourages employees to innovate continuously, and an ability to regulate LSBP activities and to establish decision-making authority in innovation projects. In addition, while other frameworks emphasize resource management, ours focuses on the entrepreneur's past activities, which will help organizations innovate their business processes and invest in the most suitable R&D.

Thirdly, the ecosystem pillar emphasizes external partner collaboration as mentioned by Lawson and Samson (2001), but also the selection of vertical and horizontal relationships, including the frequency of actors as described by Frishammar et al. (2012). Additionally, our framework emphasizes the organization's ability to screen different potential partners and the extent to which external network information influences an organization's innovation investment decisions. In sum, in accordance with our findings, the capability frameworks for general innovation and process innovation are still relevant for innovating LSBP, but mainly require enhanced and new capabilities.

5.3 Practical contributions

This study provides several practical implications for organizations because our conceptual framework can facilitate a more proficient management of process innovation in concrete LSBP practices. First, our overview of main differences and barriers can help managers and practitioners to focus their attention on the typical challenges when innovating in LSBP. Employees who are aware of these barriers and differences, for instance, might better prepare suitable solutions or workarounds. For example, "people" has been highlighted as a barrier. Innovating in LSBP requires more capability to align culture, structure, techniques, and technology because of a high dependency on knowledge workers' expertise to perform the tasks. This finding aligns with our study, indicating that one of the main capabilities to innovate in LSBP relates to employee skills. LSBP necessitates a more dynamic work profile to identify possible solutions for a variety of scenarios and to assess the best technological fit. Organizations can especially utilize our reflective questions to assess the presence of employee-related creative skills or problem-solving skills. The organization can also utilize one of the pillars for technology adoption by reflecting on their current situation to evaluate what possibilities exist for combining different technologies in order to find more creative solutions for the different process needs. Another example relates to one of the main differences identified, namely "collaboration". LSBP is associated with informal social interactions, tacit knowledge, values, and men-

tal models. Consequently, a different approach is needed. By utilizing the reflective questions, organizations might better explore the possibilities of forming collaboration agreements in vertical or horizontal alliances or executing supplier selection by function.

Moreover, practical implications reside by means of the resulting brainstorming approach (i.e., resulting from the third Delphi round). This approach can be seen as an instrument to assist organizations in reflecting on each capability to build PIC in LSBP. It helps organizations brainstorm to assess their current capability states and can therefore be used as the foundation for a gap analysis. By providing step-by-step procedures on how to conduct brainstorming, an organization should first read the differences between innovating in LSBP and SBP in order to determine the actual presence of LSBP in their organization. At this stage, it is crucial to have a person with knowledge of the process and with a management overview to assess the fit between our brainstorming purpose and their corporate needs. Once a fit has been confirmed, the organization can further identify the scope (i.e. a concrete department or process with LSBP) and continue using the reflective questions per sub-capability area. Afterwards, they can implement the necessary follow-up actions for their organization, as described in the reflective questions. However, it should be noted that our empirically validated framework is not prescriptive in nature. The framework thus acts as a guiding mechanism and does not intend to assign a specific assessment outcome (e.g., a scoring or maturity level) at this moment. Its primary aim is to provide organizations with an overview that facilitates making relevant reflections in the light of their specific business context and corporate needs. These reflections highlight that the framework is a collaborative effort between practitioners and academics, also emphasizing the need for additional research in this field.

5.4 Research limitations and future research

We acknowledge several limitations. First, although the Delphi method is known for capturing insights from experts worldwide, our research has limited African experts. Although invitations have been sent, those experts did not respond or did not agree to participate. Secondly, as is typical for Delphi studies, our results are based on the opinions of a limited panel. Nevertheless, we are confident in the validity of our findings due to the purposive composition of our panel, their positive feedback, and their high satisfaction scores. Thirdly, the definitions and capability areas have been derived from previously published research in various fields, rather than allowing the experts to formulate their own definitions as from round 1 (as opposed to a classic Delphi). However, in accordance with the exploratory nature of Delphi studies, our framework represents only one possible set of capability areas and can thus be seen as a starting point for further empirical testing in real-life case studies. Similarly, the brainstorming procedure needs to be applied in concrete organizations to better evaluate the usefulness of this artefact. Alternatively, the resulting PIC-LSBP capability framework can be used as a basis for deriving organization-specific capability development journeys (e.g., more contingent upon organization size, sector, region). It can also be used in concrete projects to analyze which capability areas can drive corporate success in different business contexts and build assessment criteria, pos-

sibly in line with a maturity model or measurement instrument. We thus encourage research to complement the explorative nature of the Delphi method.

6 Conclusion

Our study adds value to the literature of process innovation by specially bridging the gap for LSBP. The study's novelty lies in being one of the first to delve deeper into the required capabilities for PIC in LSBP and offering a practical way of assessing them in concrete organizations. Moreover, we have shed some light on the major differences and barriers between innovating SBP and LSBP (RQ1), as well as on the key success factors for innovating LSBP (RQ2). More specifically, for RQ1, we have revealed five main differences related to: (1) flexibility, (2) uncertainty, (3) empowerment, (4) collaboration, and (5) software support. In addition, we have uncovered five main barriers when innovating LSBP regarding: (1) controls, (2) outcomes, (3) technology, (4) people, (5) time and budget. Regarding RQ2, we have developed an empirically validated and refined capability framework with three pillars, six main capability areas and 18 sub-areas to build PIC in LSBP. These pillars (i.e., people-process-technology, organizational, and ecosystem-related) can help organizations better and more comprehensively achieve their performance objectives (even though future research can demonstrate the exact performance outcomes in concrete business situations). Regarding the practical implications, we have proposed a brainstorming or assessment approach that organizations can use to reflect on their current PIC in LSBP, including reflective questions per capability area and a procedure to conduct the brainstorming. More specifically, these reflective questions serve as a translation of the conceptual PIC-LSBP framework into a practical assessment overview that is more useful for organizations when brainstorming about their current innovation capabilities within their specific LSBP business context. Since this context is so varied and flexible, we deliberately avoid a prescriptive lens or a maturity model with predefined approaches. Instead, those open questions help organizations to reflect on important elements while keeping sufficient freedom to resonate to their needs.

Appendix A: Expert profiles

Table A1 Profile of practitioners

Practitioners			
IDs	Years of Experience	Industry	Current Position and Related Responsibility
R01	16	Real estate	Group Business Process Automation Manager
R02	15	Consultant	Senior Manager
R03	21	Transportation	Senior Supply Chain Management & Transformation Specialist
R04	16	Shipping	Head of Sales, Manager, Process Excellence Asia Pacific Region

Table A1 Profile of practitioners

Practitioners			
IDs	Years of Experience	Industry	Current Position and Related Responsibility
R05	14	Manufacturing	Process Improvement Material Science
R06	20	Ecommerce platform	Senior VP of Customer Excellence
R07	21	Energy	Group Director
R08	36	Consultant	BPM Architect and Industry analyst
R09	15	Energy	Global Head of Lean & Business Process Excellence Director North America
R10	19	Consultant	BPM & RPA Consultant
R11	29	Consultant	BPI Leader
R12	23	Waste management	Business Development and Innovation Manager
R13	21	Health care, Logistics consultant	Founder

Table A2 Profile of scholars

Scholars			
IDs	University's region	SSCI	Current position
R14	Europe	3	Professor Emeritus
R15	Europe	41	Professor
R16	South America	9	Professor
R17	Australasian	0	Associate Professor
R18	Europe	68	Professor
R19	Europe	25	Associate Professor
R20	Australasian	11	Associate Professor
R21	Australasian	4	Full Professor
R22	Europe	2	Associate Professor
R23	Europe	8	Assistant Professor
R24	Europe	5	Assistant Professor
R25	Australasian	2	Senior Lecturer
R26	Europe	8	Assistant Professor
R27	Australasian	6	Professor

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Data availability The author confirms that all data generated or analysed during this study are included in this published article and its supplementary information files.

Declarations

Competing interests The authors have no financial or proprietary interests in any material discussed in this article.

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